

Geometric Query: E0 vs 3D-Position Prediction

(a) E0 input query

Input to E0 route

Query:

"the trash can is the second closest to the door out of the group of cans . the trash can has a plastic bag over it ."

VLM evidence:

- candidate RGB canvases
- visual appearance and local context

(b) 3D-position input query

Input to 3D-position route

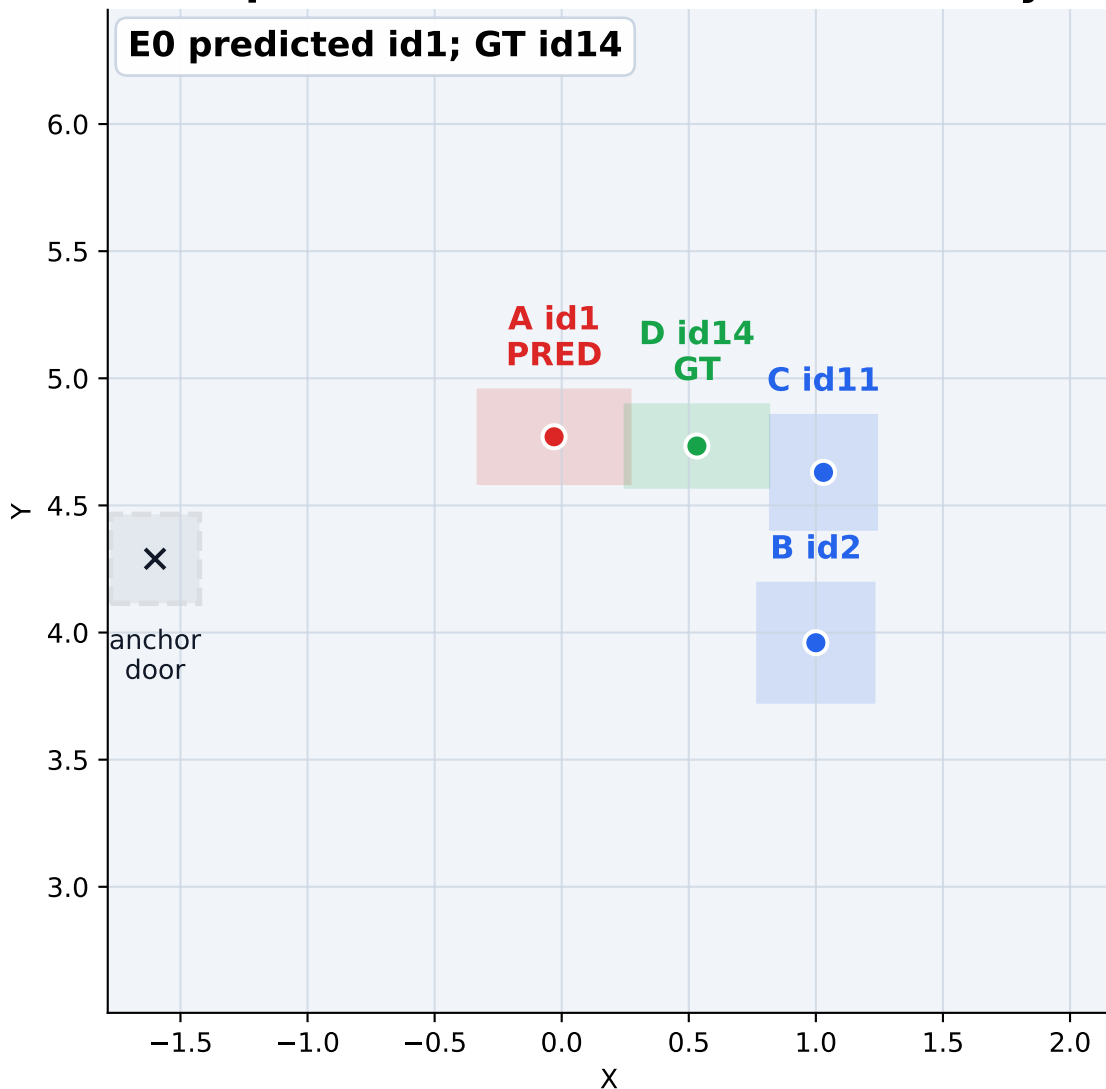
Query:

"the trash can is the second closest to the door out of the group of cans . the trash can has a plastic bag over it ."

Structured evidence:

- candidate-anchor offsets
- distances to the door anchor
- candidate sizes / heights

E0 prediction on shared BEV overlay



3D-position prediction on shared BEV overlay

